Izn Siddiqui

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PROFILE

Mechatronics engineer with field-deployed experience designing IEC-compliant embedded and protection hardware for power distribution transformers. Seeking a master's programme in electric power systems and control.

WORK EXPERIENCE

Trainee Engineer, Product and Business Development *Jan 2026 – present* | *Thal Engineering, Karachi, Pakistan*

- Designed the RS-485 communication module for the Oil Level and Transformer Sensor (OLTS), including an IEC-compliant EMI/EMC protection stage on the 240 VAC input (TVS diodes, X and Y capacitors, common-mode choke, fuse) giving surge, ESD and EFT burst protection.
- Led design of the Asset Protection Management System (APMS), an IoT-enabled LT-side transformer panel that controls a breaker based on oil level, temperature and power conditions. Specified MCCBs, magnetic contactors and enclosure components against PPMC specifications, benchmarking Chinese and German vendors on current rating, breaking capacity and price.
- Installed 2 APMS proof-of-concept panels (Dera Ghazi Khan and Lahore), in field use by distribution companies for over 4 months. Presented the system to PPMC, NTDC, MEPCO and LESCO.
- Designed DCRC (Direct Control, Remote Control) for K-Electric, a remote controller for HT-side load break switches, built as an alternative to ABB's own controller. It also trips on oil level, door opening, device failure and battery failure. Proof of concept complete, field trial ongoing.
- Led production transfer of the RS-485 module to mass production: levelled BOMs (L1 to L3), PFMEA, work instructions, control wiring diagrams and assembly jigs. Trained junior team members in soldering, wiring-diagram interpretation and PPMC compliance, and onboarded the QA team.
- Moved PCB assembly from vendor to in-house, cutting unit cost by about 25% (about PKR 10,000 per unit). Set up and trained a supply chain team for Gerber, BOM and CPL based PCB procurement.
- Secured the company's first purchase order for the module (over PKR 1 million) and prepared technical quotations for international clients in China.

Engineering Intern, R&D Department *Aug 2023* | *Yousuf Dewan Companies, Pakistan*

- Designed and implemented robotic harvesting technology that reduced labour costs by 25% and increased yield efficiency by 30%.
- Used Microsoft Excel and Power BI for data analysis.

Intern, Dataset Creation Team (remote) *Aug – Sep 2023* | *Bluediamond.AI, Salzburg, Austria*

- Created datasets for sensor instrumentation and educational chatbot design, using data analysis and programming tools to improve dataset accuracy, within an international cross-functional team.

EDUCATION AND TRAINING

Bachelor of Mechatronics Engineering

2021 – 2025 (degree awarded 6 June 2025) | National University of Sciences and Technology (NUST), Pakistan

- Relevant coursework: Linear Control Systems, Mechatronics System Design, Modeling and Simulation (MATLAB), Microcontroller and Embedded Systems, Instrumentation and Measurements, Actuating Systems, Special Topics in Mechatronics (machine learning).
- Final year project: BuildIt, Modular Robotics Kit for STEM Education (ranked Top 3 in the department).

GCE Advanced Level, The Lyceum School, Pakistan (2019 – 2021)

GCE Ordinary Level, Foundation Public School, Pakistan (2017 – 2019)

PROJECTS

Asset Protection Management System (APMS) *Industrial project, Thal Engineering*

- IoT-enabled protection and monitoring panel for LT-side transformers, combining MCCBs and magnetic contactors with a custom PCB for remote breaker control based on oil level, temperature and power conditions.

RS-485 Communication Module for Oil Level and Transformer Sensor (OLTS) *Industrial project, Thal Engineering*

- IEC-compliant module transmitting oil level and temperature data from transformer-mounted sensors to SMCD, with configurable addressing for multi-device networks.

Build-It, Modular Robotics Kit for STEM Education *Final year project, NUST -https://github.com/iznsiddiqui/Build-It*

- Designed and developed a 3D-printed modular robotics kit with building, sensor and actuator blocks, as a cheaper, localized alternative to LEGO MINDSTORMS.
- Responsible for hardware architecture selection and PCB design; developed an embedded C++ library with FreeRTOS for sensor and actuator data monitoring.
- Ranked Top 3 final year project at NUST, received multiple incubation offers, and debuted in the LEGO category at NERC 2025 against EV3 robots.

Smart Pillow for Sleep Apnea *NUST CEME Hackathon 2022 | Winner (team project)*

- IoT-enabled system with FFT snoring detection and corrective actuation that lifts the head to restore airflow.
- Responsible for the Blynk-based app and PCB design.

IoT-based Automated Hydroponics System *R&D Internship Project, Yousuf Dewan Companies*

- Developed a system for water, pH and temperature monitoring with IoT-based control for optimized crop yield.
- Responsible for microcontroller-based sensor and actuator control; built a Blynk-based app.

Autonomous Grid Following Robot *National Engineering and Robotics Contest (NERC) 2024*

- Designed an autonomous grid robot with map-based navigation for debris removal and tree planting; performed 3D simulation of the robot base in ANSYS and designed an H-bridge PCB for 12 V motors.
- Wrote the embedded code for maze navigation.

PERSONAL SKILLS

Mother tongue: Urdu

Other languages: English: IELTS Academic, overall 8.0 (C1). French: A1 (NUST French Language Level 1, qualified).

Hardware platforms: ESP32, STM32, Raspberry Pi, Arduino

Embedded and control: C, C++, Embedded C, FreeRTOS, STM32CubeMX, STM32CubeIDE, MATLAB, Simulink, PID control, ROS, NI LabVIEW

Design tools: Proteus, EasyEDA (PCB design), SolidWorks, Fusion 360, ANSYS, 3D printing

Communication protocols: RS-485, GSM/GPRS

Standards and compliance: IEC EMC and type-testing requirements, PPMC specifications

Data tools: Microsoft Excel, Power BI

Organisational and communication skills: supervised and trained a production team, benchmarked international suppliers, and presented technical systems to regulators and utility stakeholders.

EXTRACURRICULAR ACTIVITIES

- **Director, National Engineering and Robotics Contest (NERC), 2022:** part of the organising body of Pakistan's largest robotics competition, working on arenas for line-following robots and live streams.

- **President, Sports and Adventure Society, NUST CEME:** winner of the Best President Award (2025).

ADDITIONAL INFORMATION

Honours and awards: Top 3 Final Year Project, Department of Mechatronics Engineering, NUST CEME (2025); Winner, NUST CEME Hackathon (2022)

Certifications: Registered Engineer, Pakistan Engineering Council (valid until December 2027)